Projeto Final - Aprendizagem de Máquina

Modelo: TabICL v2

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Agenda

1. Resultados do Experimento

Resultados do Experimento

Tabela: Hiperparâmetros Principais

Hiperparâmetro	Tipo / faixa	Justificativa
n_estimators	int / [4, 32]	Define o tamanho do ensemble de contexto.
softmax_temperature	float / [0.3, 1.5]	Controla a calibração de confiança simulando annealing.
outlier_threshold	float / [2.0, 6.0]	Limiar para o clipping de features numéricas.
average_logits	bool / {True, False}	Estratégia de combinação estrutural do ensemble.
norm_methods	cat. / {quantile, robust, ...}	Método de escalonamento espacial do embedding contínuo.

Tabela: Espaços de Busca Optuna

Modelo	Hiperparâmetro	Tipo	Faixa de Busca
LightGBM	n_estimators	int	[50, 500]
	learning_rate	float (log)	[0.001, 0.3]
	max_depth	int	[3, 12]
	num_leaves	int	[20, 150]
	subsample	float	[0.5, 1.0]
	colsample_bytree	float	[0.5, 1.0]
XGBoost	n_estimators	int	[50, 500]
	learning_rate	float (log)	[0.001, 0.3]
	max_depth	int	[3, 12]
	subsample	float	[0.5, 1.0]
	colsample_bytree	float	[0.5, 1.0]
CatBoost	iterations	int	[50, 500]
	learning_rate	float (log)	[0.001, 0.3]
	depth	int	[3, 10]
	subsample	float	[0.5, 1.0]

Tabela: Datasets Avaliados

Dataset	TID (Task)	Instâncias	Features	Classes	Regime
blood-transfusion-service-center	1464	748	5	2	Small
diabetes	37	768	9	2	Small
anneal	2	898	39	5	Small
credit-g	168757	1.000	21	2	Medium
qsar-biodeg	359956	1.055	42	2	Medium
baseball	2077	1.340	17	3	Medium
yeast	2073	1.484	9	10	Medium
splice	45	3.190	61	3	Medium
Bioresponse	359967	3.751	1777	2	Medium
hypothyroid	3011	3.772	30	4	Medium
hiva_agnostic	3892	4.229	1618	2	Medium
spambase	43	4.601	58	2	Medium
waveform-5000	58	5.000	41	3	Medium
churn	359968	5.000	21	2	Medium
page-blocks	30	5.473	11	5	Medium
optdigits	28	5.620	65	10	Medium
satimage	2074	6.430	37	6	Medium
isolet	3481	7.797	618	26	Medium
mushroom	24	8.124	23	2	Medium
JapaneseVowels	3510	9.961	15	9	Medium
pendigits	32	10.992	17	10	Large
nursery	26	12.960	9	5	Large
letter	6	20.000	17	26	Large
houses	3688	20.640	9	2	Large
Amazon_employee_access	359979	32.769	10	2	Large
KDDCup09_appetency	3945	50.000	231	2	Large
APSFailure	168868	76.000	171	2	Large
KDD98	361329	82.318	478	2	Large
Diabetes130US	211986	101.766	50	3	Large
porto-seguro	360113	595.212	58	2	Large

Tabela: Métricas Avaliadas

Sistema	AUC-OvO	Accuracy	G-Mean	Cross-Entropy	Tempo (s)
AutoGluon Extreme	0.9108	0.9128	0.6410	0.2198	3658.0
TabICL	**0.9090**	**0.9165**	**0.6889**	**0.2025**	**48.5**
AutoGluon Default	0.9094	0.9102	0.6577	0.2320	1596.0
CatBoost Tuned	0.9002	0.9096	0.6494	0.2449	782.0
CatBoost Default	0.8989	0.9078	0.6796	0.2396	23.0
XGBoost Tuned	0.8982	0.9102	0.6553	0.2331	359.0
LightGBM Tuned	0.8977	0.9095	0.6801	0.2562	2899.0
LightGBM Default	0.8938	0.9063	0.6561	0.3237	32.0
XGBoost Default	0.8897	0.9064	0.6587	0.2688	1.0

Tabela: Análise Bayesiana com ROPE

Comparação	p(TabICL melhor)	p(equivalente)	p(pior)
TabICL vs AutoGluon extreme	0,0000	0,9991	0,0008
TabICL vs AutoGluon default	0,0121	0,9878	0,0000
TabICL vs CatBoost Tuned	0,2368	0,7631	0,0000
TabICL vs CatBoost TD	0,1353	0,8646	0,0000
TabICL vs LightGBM Tuned	0,0273	0,9726	0,0000
TabICL vs LightGBM TD	0,3076	0,6923	0,0000
TabICL vs XGBoost Tuned	0,1018	0,8982	0,0000
TabICL vs XGBoost TD	0,4276	0,5723	0,0000

Tabela: Resultados Agregados - Regime Small

Tabela 1: Resultados Agregados - Regime Small

Modelo	AUC-OVO	Acurácia	G-Mean	CE	Tempo (s)
TabICL	0.8056 ± 0.0328	0.7679 ± 0.0276	0.6273 ± 0.0405	0.4700 ± 0.0060	12.4250 ± 11.3150
LightGBM_TD	0.8423 ± 0.1418	0.8251 ± 0.1452	0.7703 ± 0.1987	0.4184 ± 0.3355	0.3033 ± 0.3782
XGBoost_TD	0.8346 ± 0.1504	0.8235 ± 0.1468	0.7578 ± 0.2086	0.4452 ± 0.3810	1.0633 ± 0.7852
CatBoost_TD	0.8313 ± 0.1565	0.8334 ± 0.1446	0.7524 ± 0.2190	0.3912 ± 0.3349	17.1100 ± 9.5178
AutoGluon_Default	0.8491 ± 0.1224	0.7705 ± 0.0140	0.4294 ± 0.3737	0.5758 ± 0.1558	1437.4433 ± 1461.1701
LightGBM_Tuned	0.8595 ± 0.1317	0.8441 ± 0.1292	0.7353 ± 0.2361	0.3403 ± 0.2603	10.8567 ± 8.6516
XGBoost_Tuned	0.8593 ± 0.1309	0.8424 ± 0.1338	0.7191 ± 0.2569	0.3286 ± 0.2676	27.2467 ± 13.6742
CatBoost_Tuned	0.8591 ± 0.1260	0.8482 ± 0.1344	0.7496 ± 0.2181	0.3385 ± 0.2817	261.1533 ± 13.2798
AutoGluon_Extreme_4h	0.8628 ± 0.1255	0.8423 ± 0.1393	0.7300 ± 0.2473	0.5076 ± 0.0584	14444.6633 ± 32.5762

Tabela: Resultados Agregados - Regime Medium

Tabela 2: Resultados Agregados - Regime Medium

Modelo	AUC-OVO	Acurácia	G-Mean	CE	Tempo (s)
TabICL	0.9425 ± 0.0740	0.9191 ± 0.1049	0.7554 ± 0.3164	0.1994 ± 0.2475	7.1865 ± 11.6278
LightGBM_TD	0.9325 ± 0.0834	0.9083 ± 0.1119	0.7385 ± 0.3154	0.2984 ± 0.3704	1.7771 ± 3.8482
XGBoost_TD	0.9222 ± 0.0923	0.9086 ± 0.1060	0.7485 ± 0.3065	0.2806 ± 0.3165	0.5782 ± 0.5632
CatBoost_TD	0.9346 ± 0.0807	0.9084 ± 0.1104	0.7447 ± 0.3100	0.2407 ± 0.2776	14.2624 ± 8.7791
AutoGluon_Default	0.9248 ± 0.0878	0.9000 ± 0.1227	0.6730 ± 0.3960	0.2512 ± 0.2747	1349.1765 ± 1095.7133
LightGBM_Tuned	0.9241 ± 0.0892	0.9114 ± 0.1081	0.7472 ± 0.3121	0.2668 ± 0.2589	178.5612 ± 436.9737
XGBoost_Tuned	0.9306 ± 0.0784	0.9105 ± 0.1088	0.7520 ± 0.3067	0.2414 ± 0.2885	91.1965 ± 118.5494
CatBoost_Tuned	0.9359 ± 0.0766	0.9091 ± 0.1058	0.7442 ± 0.3075	0.2348 ± 0.2491	323.0129 ± 174.2245
AutoGluon_Extreme_4h	0.9453 ± 0.0699	0.9157 ± 0.1009	0.7557 ± 0.3102	0.2878 ± 0.3888	14464.2382 ± 57.8431

Tabela 3: Resultados Agregados - Regime Large

Modelo	AUC-OVO	Acurácia	G-Mean	CE	Tempo (s)
TabICL	0.8629 ± 0.1616	0.9357 ± 0.1186	0.5585 ± 0.4618	0.1677 ± 0.2593	125.8050 ± 276.4557
LightGBM_TD	0.8420 ± 0.1538	0.9284 ± 0.1159	0.4555 ± 0.4377	0.2254 ± 0.2524	1.6600 ± 1.5926
XGBoost_TD	0.8418 ± 0.1676	0.9298 ± 0.1183	0.4711 ± 0.4300	0.1911 ± 0.2542	0.8290 ± 0.7095
CatBoost_TD	0.8532 ± 0.1560	0.9310 ± 0.1184	0.4378 ± 0.4528	0.1834 ± 0.2560	13.7000 ± 11.4826
AutoGluon_Default	0.8337 ± 0.1536	0.9282 ± 0.1171	0.3913 ± 0.4183	0.1959 ± 0.2503	1525.3289 ± 847.4121
LightGBM_Tuned	0.8554 ± 0.1550	0.9322 ± 0.1174	0.4708 ± 0.4399	0.1834 ± 0.2515	222.0350 ± 257.4562
XGBoost_Tuned	0.8454 ± 0.1522	0.9246 ± 0.1149	0.5164 ± 0.4187	0.2019 ± 0.2450	130.3811 ± 125.5904
CatBoost_Tuned	0.8555 ± 0.1556	0.9318 ± 0.1179	0.4600 ± 0.4469	0.2101 ± 0.2525	279.7610 ± 57.8396
AutoGluon_Extreme_4h	0.8639 ± 0.1577	0.9268 ± 0.1300	0.3081 ± 0.4665	0.3305 ± 0.3777	14912.5790 ± 585.6079

Tabela: Resultados Agregados - Datasets Binário

Tabela 4: Resultados Agregados - Datasets Binário

Modelo	AUC-OVO	Acurácia	G-Mean	CE	Tempo (s)
TabICL	0.8562 ± 0.1243	0.9087 ± 0.0933	0.6146 ± 0.3615	0.2233 ± 0.1843	87.2573 ± 228.8566
LightGBM_TD	0.8419 ± 0.1279	0.9014 ± 0.1002	0.6199 ± 0.3416	0.2668 ± 0.2253	0.7027 ± 0.7069
XGBoost_TD	0.8322 ± 0.1372	0.8993 ± 0.1017	0.6259 ± 0.3318	0.2907 ± 0.2539	0.5753 ± 0.5707
CatBoost_TD	0.8422 ± 0.1280	0.9020 ± 0.0998	0.6025 ± 0.3605	0.2525 ± 0.2168	21.1720 ± 8.4415
AutoGluon_Default	0.8286 ± 0.1221	0.8882 ± 0.1089	0.4788 ± 0.3893	0.2646 ± 0.2018	1430.8153 ± 1060.2139
LightGBM_Tuned	0.8479 ± 0.1243	0.9063 ± 0.0933	0.6226 ± 0.3379	0.2549 ± 0.1929	62.4800 ± 69.8540
XGBoost_Tuned	0.8538 ± 0.1234	0.9050 ± 0.0943	0.6132 ± 0.3482	0.2320 ± 0.1832	43.4647 ± 35.8443
CatBoost_Tuned	0.8503 ± 0.1217	0.9067 ± 0.0924	0.6197 ± 0.3445	0.2513 ± 0.1854	266.5800 ± 35.0311
AutoGluon_Extreme_4h	0.8592 ± 0.1206	0.9068 ± 0.0919	0.5348 ± 0.3993	0.2495 ± 0.1772	14649.7427 ± 509.0847

Tabela: Resultados Agregados - Datasets Multiclasse

Tabela 5: Resultados Agregados - Datasets Multiclasse

Modelo	AUC-OVO	Acurácia	G-Mean	CE	Tempo (s)
TabICL	0.9483 ± 0.0909	0.9103 ± 0.1339	0.7393 ± 0.3616	0.2085 ± 0.3082	7.2423 ± 10.7894
LightGBM_TD	0.9446 ± 0.0913	0.9120 ± 0.1327	0.6748 ± 0.4047	0.3052 ± 0.4109	2.4787 ± 4.0971
XGBoost_TD	0.9410 ± 0.0971	0.9150 ± 0.1277	0.6880 ± 0.3966	0.2438 ± 0.3497	0.8453 ± 0.6836
CatBoost_TD	0.9522 ± 0.0879	0.9149 ± 0.1321	0.6838 ± 0.4002	0.2208 ± 0.3254	7.5473 ± 4.1262
AutoGluon_Default	0.9451 ± 0.0895	0.9047 ± 0.1348	0.6306 ± 0.4327	0.2658 ± 0.3370	1402.6260 ± 1014.0618
LightGBM_Tuned	0.9415 ± 0.0964	0.9169 ± 0.1309	0.6851 ± 0.4042	0.2378 ± 0.3079	290.0840 ± 485.3656
XGBoost_Tuned	0.9363 ± 0.0953	0.9118 ± 0.1296	0.7272 ± 0.3563	0.2420 ± 0.3362	152.2614 ± 143.1085
CatBoost_Tuned	0.9526 ± 0.0865	0.9145 ± 0.1300	0.6803 ± 0.4016	0.2225 ± 0.3019	338.2393 ± 184.7561
AutoGluon_Extreme_4h	0.9606 ± 0.0852	0.9173 ± 0.1341	0.6731 ± 0.4275	0.3985 ± 0.4801	14573.7127 ± 243.3457

Tabela: Resultados Agregados - Maioria Categórica

Tabela 6: Resultados Agregados - Maioria Categórica

Modelo	AUC-OVO	Acurácia	G-Mean	CE	Tempo (s)
TabICL	0.8571 ± 0.1414	0.8907 ± 0.1395	0.5491 ± 0.4021	0.2459 ± 0.2909	105.3672 ± 293.6118
LightGBM_TD	0.8553 ± 0.1426	0.9083 ± 0.1377	0.4631 ± 0.4351	0.2662 ± 0.2927	0.7622 ± 0.5561
XGBoost_TD	0.8414 ± 0.1492	0.9094 ± 0.1425	0.4752 ± 0.4291	0.2283 ± 0.3221	0.4589 ± 0.2953
CatBoost_TD	0.8642 ± 0.1471	0.9122 ± 0.1398	0.4550 ± 0.4399	0.2139 ± 0.3022	12.1889 ± 8.9086
AutoGluon_Default	0.8531 ± 0.1372	0.8880 ± 0.1425	0.3563 ± 0.4111	0.2973 ± 0.3232	1136.1989 ± 866.4667
LightGBM_Tuned	0.8471 ± 0.1424	0.9140 ± 0.1363	0.4833 ± 0.4332	0.2248 ± 0.2877	79.9856 ± 84.3158
XGBoost_Tuned	0.8457 ± 0.1337	0.9063 ± 0.1344	0.5371 ± 0.3959	0.2237 ± 0.2780	65.3035 ± 47.5822
CatBoost_Tuned	0.8694 ± 0.1443	0.9133 ± 0.1383	0.4756 ± 0.4333	0.2096 ± 0.2887	255.9011 ± 27.3809
AutoGluon_Extreme_4h	0.8830 ± 0.1471	0.9079 ± 0.1455	0.4110 ± 0.4710	0.3363 ± 0.2934	14761.0400 ± 642.9783

Tabela: Resultados Agregados - Maioria Numérica

Tabela 7: Resultados Agregados - Maioria Numérica

Modelo	AUC-OVO	Acurácia	G-Mean	CE	Tempo (s)
TabICL	0.9216 ± 0.1024	0.9176 ± 0.1031	0.7317 ± 0.3371	0.2031 ± 0.2365	22.3424 ± 49.9065
LightGBM_TD	0.9095 ± 0.1103	0.9060 ± 0.1087	0.7263 ± 0.3161	0.2945 ± 0.3461	1.9457 ± 3.5662
XGBoost_TD	0.9060 ± 0.1183	0.9062 ± 0.1030	0.7348 ± 0.3062	0.2840 ± 0.2984	0.8181 ± 0.7120
CatBoost_TD	0.9113 ± 0.1098	0.9069 ± 0.1069	0.7238 ± 0.3247	0.2464 ± 0.2656	15.2900 ± 9.8304
AutoGluon_Default	0.9013 ± 0.1134	0.9000 ± 0.1138	0.6398 ± 0.3910	0.2515 ± 0.2561	1536.9442 ± 1075.2053
LightGBM_Tuned	0.9151 ± 0.1051	0.9106 ± 0.1034	0.7269 ± 0.3193	0.2556 ± 0.2431	217.5519 ± 423.0166
XGBoost_Tuned	0.9162 ± 0.1041	0.9093 ± 0.1038	0.7272 ± 0.3233	0.2427 ± 0.2676	111.8171 ± 134.4881
CatBoost_Tuned	0.9152 ± 0.1027	0.9095 ± 0.1009	0.7247 ± 0.3205	0.2486 ± 0.2331	322.3419 ± 158.0779
AutoGluon_Extreme_4h	0.9215 ± 0.1000	0.9138 ± 0.1003	0.6867 ± 0.3661	0.3188 ± 0.3964	14547.7367 ± 212.9233

Tabela 8: Resultados Agregados - Com Valores Ausentes (NaNs)

Modelo	AUC-OVO	Acurácia	G-Mean	CE	Tempo (s)
TabICL	0.8479 ± 0.1519	0.9417 ± 0.0802	0.3887 ± 0.4366	0.1515 ± 0.1727	152.7031 ± 306.7034
LightGBM_TD	0.8677 ± 0.1585	0.9683 ± 0.0404	0.4424 ± 0.4535	0.1227 ± 0.1387	0.9537 ± 0.7491
XGBoost_TD	0.8335 ± 0.1684	0.9691 ± 0.0408	0.4578 ± 0.4508	0.1099 ± 0.1290	0.6687 ± 0.4765
CatBoost_TD	0.8720 ± 0.1529	0.9694 ± 0.0413	0.4336 ± 0.4685	0.0978 ± 0.1213	11.5337 ± 10.8205
AutoGluon_Default	0.8434 ± 0.1571	0.9415 ± 0.0818	0.3157 ± 0.4451	0.1947 ± 0.2551	1695.8725 ± 1144.7099
LightGBM_Tuned	0.8489 ± 0.1545	0.9699 ± 0.0392	0.4562 ± 0.4623	0.1101 ± 0.1118	91.4100 ± 83.0717
XGBoost_Tuned	0.8515 ± 0.1525	0.9708 ± 0.0399	0.4492 ± 0.4748	0.1038 ± 0.1216	64.2450 ± 39.3803
CatBoost_Tuned	0.8695 ± 0.1527	0.9698 ± 0.0410	0.4387 ± 0.4662	0.1294 ± 0.1421	265.2950 ± 36.1851
AutoGluon_Extreme_4h	0.8763 ± 0.1566	0.9653 ± 0.0383	0.3181 ± 0.4608	0.2079 ± 0.1930	14789.6100 ± 672.4407

Tabela: Resultados Agregados - Sem Valores Ausentes

Tabela 9: Resultados Agregados - Sem Valores Ausentes

Modelo	AUC-OVO	Acurácia	G-Mean	CE	Tempo (s)
TabICL	0.9221 ± 0.0980	0.8978 ± 0.1228	0.7818 ± 0.2704	0.2393 ± 0.2720	8.9032 ± 12.1903
LightGBM_TD	0.9026 ± 0.1073	0.8843 ± 0.1263	0.7219 ± 0.3126	0.3454 ± 0.3559	1.8223 ± 3.5004
XGBoost_TD	0.9060 ± 0.1102	0.8846 ± 0.1237	0.7294 ± 0.3021	0.3245 ± 0.3269	0.7255 ± 0.6915
CatBoost_TD	0.9063 ± 0.1108	0.8863 ± 0.1258	0.7194 ± 0.3159	0.2871 ± 0.2955	15.3873 ± 9.0541
AutoGluon_Default	0.9027 ± 0.1045	0.8801 ± 0.1297	0.6416 ± 0.3719	0.2909 ± 0.2804	1315.2109 ± 978.7403
LightGBM_Tuned	0.9114 ± 0.1028	0.8904 ± 0.1224	0.7257 ± 0.3080	0.2959 ± 0.2726	207.1445 ± 415.6504
XGBoost_Tuned	0.9109 ± 0.0994	0.8857 ± 0.1208	0.7505 ± 0.2638	0.2854 ± 0.2892	110.0878 ± 132.7904
CatBoost_Tuned	0.9131 ± 0.1014	0.8891 ± 0.1208	0.7268 ± 0.3038	0.2760 ± 0.2669	315.9059 ± 155.7452
AutoGluon_Extreme_4h	0.9221 ± 0.0972	0.8927 ± 0.1253	0.7079 ± 0.3489	0.3662 ± 0.4039	14547.0432 ± 217.1583

Gráficos: Diagramas de Diferença Crítica (AUC e ACC)

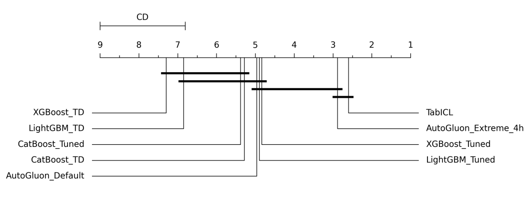


Figura 1: CD Diagram: AUC-OvO

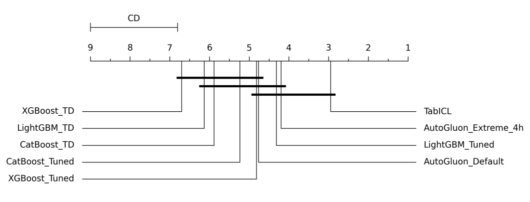


Figura 2: CD Diagram: Acurácia

Gráficos: Diagramas de Diferença Crítica (G-Mean e CE)

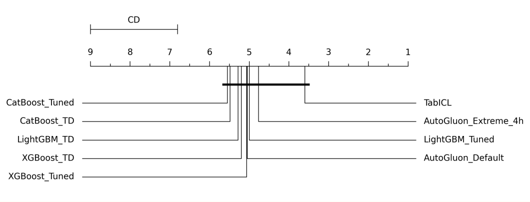


Figura 3: CD Diagram: G-Mean

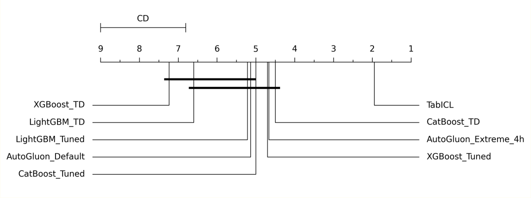
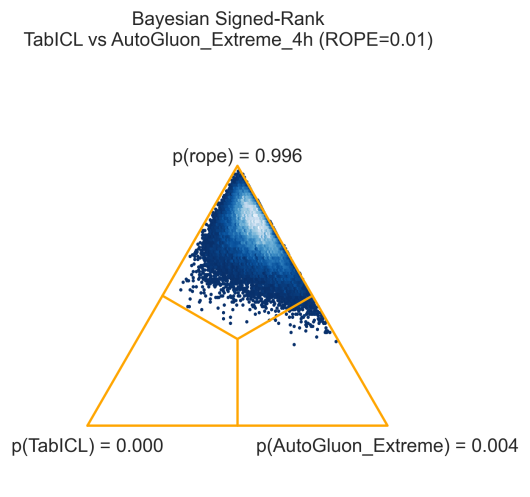
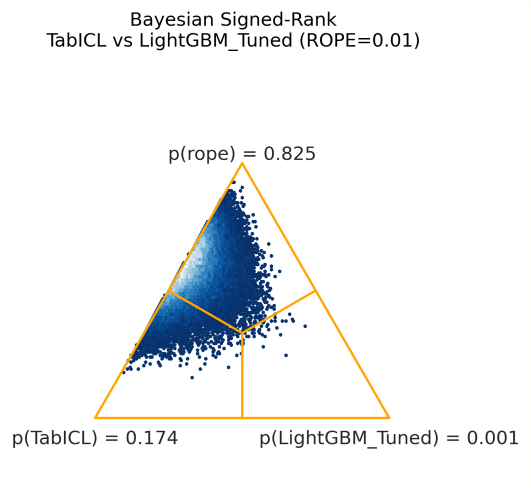


Figura 4: CD Diagram: Cross-Entropy

Gráficos: Análise Bayesiana (Plots Simplex)





Gráficos: Custo vs Desempenho e Regime

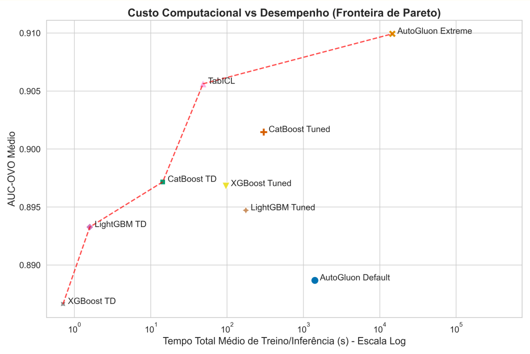


Figura 7: Gráfico de Dispersão Pareto

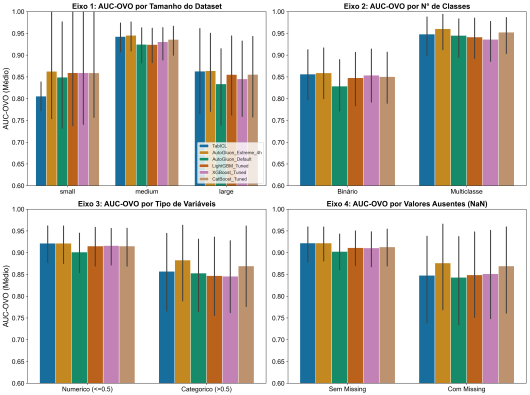


Figura 8: Desempenho por Regime

Gráficos: Otimização Optuna (Small)

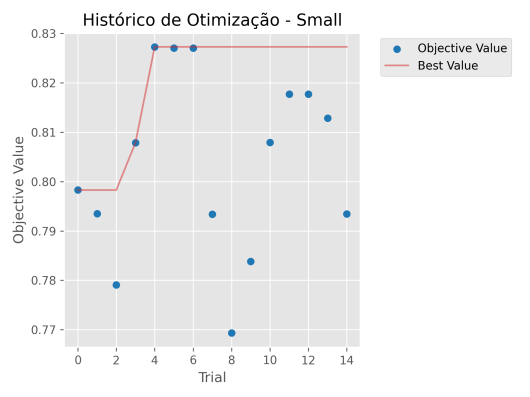


Figura 9: Optuna History (Small)

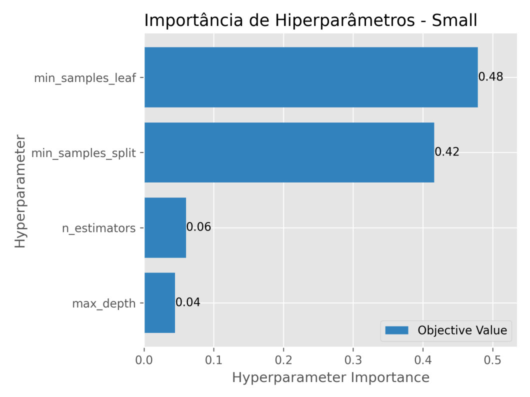


Figura 10: Optuna Importances (Small)

Gráficos: Otimização Optuna (Medium)

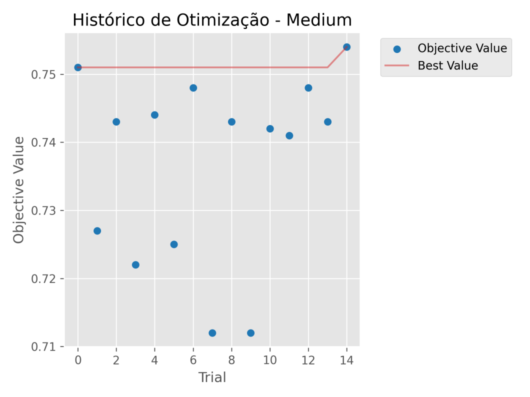


Figura 11: Optuna History (Medium)

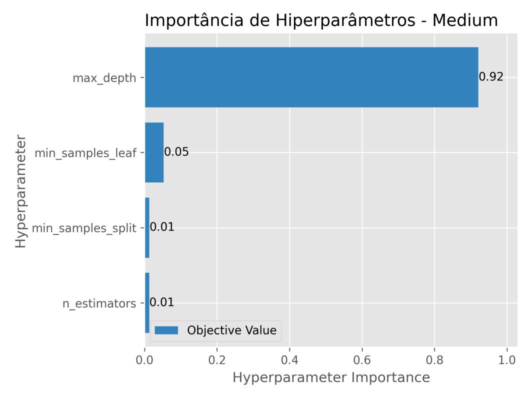


Figura 12: Optuna Importances (Medium)

Gráficos: Otimização Optuna (Large)

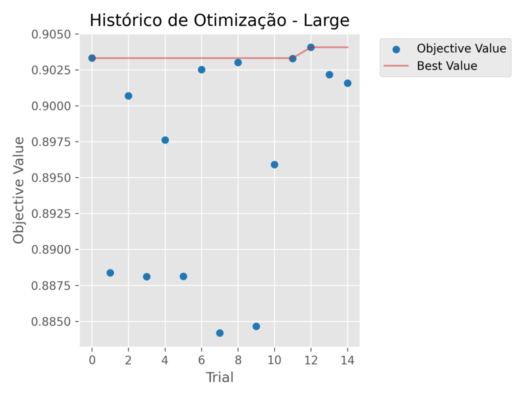


Figura 13: Optuna History (Large)

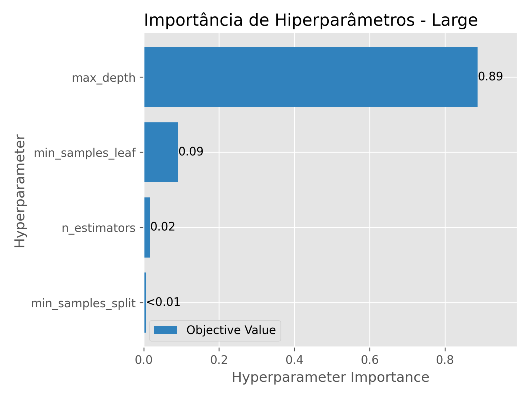


Figura 14: Optuna Importances (Large)

Agradecemos pela Atenção!

Repositório: [github.com/grupo/projeto-am]
Referências principais: Qu et al. (2025/2026), Hölzmuller (2024), Demšar (2006).

Dúvidas ou Questionamentos?